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Transfert learning



1. Abstract / What did I do in a nutshell?

We've build a basic CNN model based upon transfer learning.

2. Introduction / What is the problem?

We have a dataset (CIFAR 10) containing 32x32 pixel images, divided into 10 categories, and we want to train a CNN model that can successfully (minimum accuracy of 87%) predict the categories for similar pictures.

3. Materials and Methods / How did I solve the problem?

We could build a model from scratch, but the dataset, even though it looks big, is not really sufficient to do so, and there's also the advantage of efficiency by using already trained models, adjusting the output layer.

4. Results / What did I find out?

MobileNetV2 was choses as the base model, basically because of its relative small size and the computing speed. We used ImageNet pretrained weights, and added a small

classification head on top. We reached a validation accuracy of 87.45% on the CIFAR-10 test set, making it to the 87% threshold. Training took approximately 5 minutes on CPU: around 2,5 minutes for the one-time feature extraction, and another 2,5 minutes for 20 epochs of head training. Computing the features once upfront, as suggested in the assignment - rather than at every epoch - was key to keeping training fast. We tested some dropout rates, and landed on 0.5, and early stopping was planned in the model, but did not kick in.

5. **Discussion / What does it mean?**

Transfer learning works quite good, and trains fast. In this assignment at least.

6. **Literature Cited / Whose work did I refer to?**

- Holberton transfer learning project (https://intranet.hbtn.io/projects/2311)

- Keras Applications documentation : https://keras.io/api/applications/